

Business Requirements Document

PPE Compliance Detection System

Computer Vision — Object Detection Project

Field	Detail
Document Version	0.1 (Draft — for review with Product Owner)
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Status	DRAFT — items marked [TBD] require input from Business Owner or joint scoping session

1. Business Context

1.1 Problem Statement

Construction and industrial sites require workers to wear Personal Protective Equipment (PPE) such as hardhats, safety vests, gloves, goggles, and masks at all times. Today, compliance is monitored manually by safety officers walking the site or reviewing CCTV footage after the fact. This is labour-intensive, inconsistent, and reactive — violations and fall incidents are often identified only after an accident has occurred or during a compliance audit.

1.2 Proposed Solution

A computer vision system that connects to existing site CCTV cameras, automatically detects workers and their PPE status in real time, and raises an alert the moment a violation (e.g., no hardhat, no safety vest) or a fall is detected. This shifts safety monitoring from manual and reactive to automated and proactive.

1.3 Business Value

- Reduce workplace injuries and fatalities caused by missing PPE.
- Reduce regulatory/compliance risk and potential fines.
- Reduce dependency on manual, walk-the-floor safety monitoring.
- Create an auditable log of compliance for insurance and regulatory reporting.
- Enable faster emergency response through automated fall detection.

2. Objectives & Success Metrics

2.1 Business Objectives

- Automatically detect PPE compliance and non-compliance from live camera feeds.
- Notify the right people instantly when a violation or fall is detected.
- Maintain a historical, searchable log of all detected events for reporting and audits.

2.2 Success Metrics (to be finalised with Product Owner)

Metric	Target [TBD]	Notes
Detection accuracy (mAP)	[TBD — e.g., $\geq 90\%$]	Measured against held-out test set (4,423 images available).
False alert rate	[TBD — e.g., $< 5\%$]	Too many false alerts will cause the system to be ignored by site staff.
Alert latency	[TBD — e.g., < 5 seconds]	Time from violation occurring on camera to alert reaching the responsible person.
Reduction in manual safety inspection effort	[TBD]	Business-level KPI, measured post-launch.
System uptime	[TBD — e.g., 99%]	Relevant since this is a safety-critical monitoring tool.

3. Scope

3.1 In Scope

Detection classes, based on the finalised training dataset:

Category	Classes
People & Equipment	Person, Ladder, Safety Cone
PPE Present	Hardhat, Safety Vest, Gloves, Goggles, Mask
PPE Violations	NO-Hardhat, NO-Safety Vest, NO-Gloves, NO-Goggles, NO-Mask
Incident	Fall-Detected

- Live CCTV camera feed ingestion and real-time detection.
- Instant alerting when a violation or fall is detected.
- Logging of all detection events for later review and reporting.
- A dashboard for safety officers/managers to view live feeds, alerts, and history.

3.2 Out of Scope (initial phase)

- Identifying or tracking specific individual workers (facial recognition / identity).
- Assessing PPE quality or damage (e.g., a cracked hardhat is still counted as "Hardhat").
- Automated disciplinary action — the system flags and notifies; it does not take action against workers.
- Camera hardware procurement/installation — [confirm with Business Owner whether this is in scope].

3.3 Deployment Model

Confirmed: the system will process live CCTV/camera feeds in real time (not just uploaded video/image review).

Scale (number of sites, number of cameras, concurrent feeds) is not yet finalised and should be agreed in the first scoping session with the delivery team, as it directly drives infrastructure cost and architecture. [TBD]

4. Users & Stakeholders

Role	Interaction with the System
Safety Officer	Primary user. Monitors live dashboard, receives instant alerts, reviews violation history.
Site Manager	Receives summarised alerts/reports, uses data for team briefings and corrective action.
Compliance / Legal	Uses historical logs and reports for audits and regulatory reporting.
System Administrator	Configures cameras, users, alert rules, and access permissions.
Executive Sponsor	Reviews high-level compliance trends and ROI (e.g., violation trend dashboard).

5. Functional Requirements

5.1 Backend / Detection Engine

- Ingest live video streams from connected CCTV cameras (protocol e.g., RTSP — to confirm with existing camera infrastructure).
- Run the trained object detection model on incoming frames to identify all 14 classes listed in Section 3.1.
- Apply a configurable confidence threshold to reduce false positives.
- Generate a structured detection event for every violation/fall, including: camera ID, site/location, timestamp, class detected, confidence score, and snapshot image with bounding box.
- Trigger the alerting service immediately when a violation or fall event is generated.
- Store all detection events in a database for historical search and reporting.
- Expose an API for the frontend dashboard (and any future third-party integrations) to query live status and historical events.
- Support user authentication and role-based access control (Safety Officer, Manager, Admin, Compliance).

5.2 Alerting

- Instant alert on violation/fall detection — channel(s) to confirm: dashboard pop-up, email, SMS, or a combination. [TBD]
- Alert should include: what was detected, which camera/location, timestamp, and a snapshot image.
- Configurable alert routing — e.g., fall detection may need to notify a different escalation path than a missing-glove detection. [TBD — confirm severity tiers with Business Owner].
- Every alert is also logged, even if the recipient does not act on it, for audit purposes.

5.3 Frontend / Dashboard

- Live view: real-time feed(s) with detection overlays (bounding boxes and labels) for selected cameras.
- Alerts panel: real-time feed of active/recent violations, with the ability to acknowledge or dismiss.
- Historical log: searchable and filterable by site, camera, date range, violation type, and worker area.
- Reporting: summary charts (e.g., violations by type over time, by site, by camera) exportable for compliance reporting.
- Camera/site management screen (Admin): add/remove cameras, assign to sites, configure alert rules.
- User management screen (Admin): manage roles and permissions.
- Responsive design so Safety Officers can check alerts from a tablet/phone while on-site. [confirm device requirements]

5.4 Data & Storage

- Store detection event metadata (class, confidence, timestamp, camera, site) indefinitely or per retention policy [TBD].
- Store snapshot images (not necessarily full video) tied to each violation event.
- Define footage/image retention period — this has privacy and storage-cost implications and should be confirmed with legal/compliance. [TBD]
- Training/validation/test dataset (30,765 / 8,814 / 4,423 images respectively) is available and documented in the accompanying DATASET.md for the ML team.

6. Non-Functional Requirements

Category	Requirement
Performance	Real-time detection with alert latency target of [TBD] seconds from event to notification.
Scalability	Architecture should support adding more cameras/sites over time without redesign. Initial scale [TBD].
Reliability	System should target [TBD]% uptime given its safety-critical nature.
Accuracy	Minimise false positives to avoid alert fatigue; minimise false negatives to avoid missed safety incidents.
Security & Privacy	Worker footage is sensitive personal data. Access must be role-restricted; retention and access policy must align with local labour/privacy law.
Usability	Dashboard must be usable by non-technical safety staff with minimal training.
Integration	Should integrate with existing CCTV/NVR systems where feasible; API should allow future integration with other safety software.

7. Supporting Data Reference

A labelled object detection dataset is already available for model training (YOLOv8 format), documented in full in the attached DATASET.md. Summary:

Split	Images	Labels
Train	30,765	30,765
Validation	8,814	8,814
Test	4,423	4,423

Detected classes: Fall-Detected, Gloves, Goggles, Hardhat, Ladder, Mask, NO-Gloves, NO-Goggles, NO-Hardhat, NO-Mask, NO-Safety Vest, Person, Safety Cone, Safety Vest.

Note for engineering team: the class "NO-Safety Vest" contains a space and should be handled carefully in code/config to avoid parsing issues.

8. Assumptions & Constraints

- CCTV camera hardware is assumed to already exist on site; procurement/installation scope to be confirmed. [TBD]
- Number of sites and cameras is not yet finalised; to be agreed with the delivery team before technical design. [TBD]

- Budget ceiling and target timeline are not yet defined in this draft. [TBD]
- Hosting environment (cloud provider vs on-premise, given real-time video processing needs) is not yet decided. [TBD]
- Existing IT/network infrastructure must support live video streaming to the detection system; to be validated during technical discovery.

9. Acceptance Criteria (Draft)

- When a worker without a hardhat is visible on a live camera feed, the system detects it and raises a dashboard alert within [TBD] seconds.
- The relevant Safety Officer/Manager receives a notification through the agreed channel(s) for every violation and fall event.
- Every detection event (violation or fall) is stored and retrievable through the historical log and reporting screens.
- Only authorised users (per role) can access live feeds, historical footage, and admin settings.
- The system distinguishes between the 14 defined classes with an accuracy level agreed in Section 2.2.

10. Open Items for Kickoff Discussion

- Final scale: number of sites and cameras for Phase 1 (pilot vs full rollout).
- Alert channels and escalation rules per violation severity (e.g., fall vs missing gloves).
- Footage/image retention period and privacy sign-off.
- Camera hardware/network readiness at each site.
- Budget and target go-live date.
- Success metric targets (accuracy, false alert rate, latency, uptime).